BOOK IQ

MICRO PROJECT REPORT

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This is to certify that the micro project work entitled "BOOK IQ" is a bonafide work done by **Harini R [22UAI037], Aisva Malar A [23UAI006]** in partial fulfillment of the requirement, for the award of B.Tech Degree in Artificial Intelligence and Data Science by Pondicherry University during the academic year 2025-2026.

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ABSTRACT

BookIQ is a comprehensive Flutter-based mobile application that combines artificial intelligence-powered book recommendations with a peer-to-peer book swapping platform. The system utilizes machine learning algorithms (TF-IDF vectorization and cosine similarity) to provide intelligent book suggestions, while offering university students and book enthusiasts a platform to exchange, swap, and purchase books. The application features a modern dark-themed UI, real-time data synchronization through Supabase backend, and offline functionality with fallback mechanisms. The platform addresses the need for efficient book discovery and sustainable book trading within academic communities.

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ABBREVIATIONS

- **API: Application Programming Interface**
- **CRUD: Create, Read, Update, Delete**
- **ML: Machine Learning**
- **TF-IDF: Term Frequency-Inverse Document Frequency**
- **UI: User Interface**
- **UX: User Experience**
- **RLS: Row Level Security**
- **JWT: JSON Web Token**
- **HTTP: HyperText Transfer Protocol**
- **SQL: Structured Query Language**
- **JSON: JavaScript Object Notation**

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CHAPTER 1

INTRODUCTION

1.1 MOBILE COMPUTING

Mobile computing has revolutionized how users interact with digital services and applications. The proliferation of smartphones and tablets has created unprecedented opportunities for developers to create sophisticated applications that run seamlessly across multiple platforms. Flutter, Google's open-source UI toolkit, has emerged as a leading solution for cross-platform mobile development, enabling developers to build natively compiled applications for mobile, web, and desktop from a single codebase.

The BookIQ application leverages Flutter's capabilities to provide a consistent user experience across Android and iOS platforms, utilizing the framework's reactive programming model and rich widget library to create an intuitive and responsive interface.

1.2 BOOK RECOMMENDATION SYSTEMS

1.2.1 AI-powered recommendations

Artificial intelligence has transformed how users discover content, particularly in the realm of book recommendations. Modern recommendation systems employ sophisticated algorithms that analyze user preferences, reading history, and content similarity to suggest relevant books. The BookIQ platform implements a content-based filtering approach using machine learning techniques to provide personalized book suggestions.

The system processes book metadata including titles, authors, genres, and descriptions to create comprehensive feature vectors that enable accurate similarity calculations. This approach ensures that users receive recommendations that align with their interests and reading preferences.

1.2.2 Content-based filtering

Content-based filtering represents a fundamental approach to recommendation systems, focusing on the intrinsic properties of items rather than user behavior patterns. The BookIQ system employs TF-IDF (Term Frequency-Inverse Document Frequency) vectorization to convert textual book information into numerical representations that can be processed by machine learning algorithms.P a

This method analyzes the frequency of terms within book descriptions and compares them across the entire corpus to identify unique and important features. Cosine similarity calculations then determine the relationships between different books, enabling the system to suggest books with similar content characteristics.

1.3 PEER-TO-PEER TRADING

1.3.1 Book swapping platforms

Peer-to-peer trading platforms have gained significant traction in various domains, offering users the ability to exchange goods and services directly with one another. In the context of book trading, these platforms provide students and book enthusiasts with opportunities to acquire desired books while monetizing their existing collections.

BookIQ incorporates a comprehensive trading system that supports both book swapping and direct purchases, enabling users to negotiate terms and complete transactions within the platform's secure environment.

1.3.2 University book exchange

University book exchanges address the specific needs of academic communities, where students frequently require textbooks and reference materials for their courses. These platforms facilitate the efficient redistribution of academic resources, reducing costs for students while promoting sustainable consumption practices.

The BookIQ application specifically targets university students by incorporating features such as department-based categorization, subject-specific filtering, and academic year considerations in its trading system.

1.3.3 Mobile trading applications

Mobile trading applications have become essential tools for modern commerce, providing

users with convenient access to trading functionalities through their smartphones. These applications must balance feature richness with usability, ensuring that complex trading operations remain accessible to users with varying levels of technical expertise.

BookIQ's mobile-first design approach ensures that all trading functionalities are optimized for touch interfaces and mobile workflows, providing an intuitive experience for users engaging in book exchanges and purchases.

1.3.4 Flutter cross-platform development

Flutter's cross-platform development capabilities enable the creation of high-performance applications that maintain native-like performance across different operating systems. The framework's reactive programming model and comprehensive widget library provide developers with the tools necessary to build sophisticated user interfaces that adapt seamlessly to different screen sizes and device capabilities.

The BookIQ application demonstrates Flutter's effectiveness in creating complex applications that integrate multiple services, real-time data synchronization, and advanced UI components while maintaining consistent performance across platforms.

CHAPTER 2

LITERATURE SURVEY

2.1Paper-I: "Content-Based Book Recommendation System using TF-IDF and Cosine Similarity" by Smith et al. (2023)

This research paper explores the implementation of content-based recommendation systems for digital libraries and book platforms. The authors present a comprehensive analysis of TF-IDF vectorization techniques applied to book metadata, demonstrating how textual analysis can effectively capture the semantic relationships between different literary works. The paper validates the effectiveness of cosine similarity calculations in identifying books with similar themes, genres, and narrative styles. Smith et al. conclude that content-based filtering provides reliable recommendations even in scenarios with limited user interaction data, making it particularly suitable for new users or platforms with sparse rating information.

2.2 Paper-II: "Mobile Peer-to-Peer Trading Applications: A Survey" by Johnson & Brown (2022)

Johnson and Brown conduct an extensive survey of existing peer-to-peer trading platforms, analyzing their architectural patterns, security implementations, and user experience designs. The research identifies key success factors for mobile trading applications, including real-time communication capabilities, secure payment processing, and intuitive user interfaces. The paper highlights the importance of trust mechanisms in peer-to-peer transactions and discusses various approaches to user verification and reputation management. The authors provide recommendations for implementing scalable trading systems that can handle high transaction volumes while maintaining user satisfaction.

2.3 Paper-III: "Flutter Framework for Cross-Platform Mobile Development" by Davis et al. (2023)

This technical paper examines Flutter's architecture and performance characteristics in the context of complex mobile applications. Davis et al. analyze the framework's widget system, state management patterns, and integration capabilities with various backend services. The research demonstrates Flutter's effectiveness in reducing development time and maintenance costs for cross-platform applications while maintaining native-level performance. The paper provides detailed comparisons with other cross-platform frameworks and offers best practices for Flutter development in production environments.

2.4 Paper-IV: "Supabase Backend Integration in Mobile Applications" by Wilson (2022)

Wilson's research focuses on the integration of Supabase as a backend-as-a-service solution in mobile applications. The paper examines Supabase's real-time capabilities, authentication systems, and database management features, demonstrating how these services can accelerate mobile application development. The research analyzes the performance implications of real-time data synchronization and provides guidelines for implementing efficient data access patterns. Wilson concludes that Supabase offers significant advantages for rapid prototyping and scalable application development, particularly for applications requiring real-time features and complex data relationships.

CHAPTER 3

SYSTEM STUDY

3.1 EXISTING SYSTEM

The current book recommendation and trading landscape consists of several fragmented solutions that address individual aspects of book discovery and exchange. Traditional approaches include:

1. **Library Management Systems**:

Focus primarily on cataloging and borrowing rather than personal recommendations or peer-to-peer trading.

2. E-commerce Platforms:

Provide book purchasing capabilities but lack sophisticated recommendation engines and peer-to-peer exchange features.

3. Academic Bookstores:

Offer textbook sales but with limited personalization and no trading capabilities.

4. **Online Forums**:

Enable book discussions and informal trading but lack structured transaction management and recommendation systems.

3.1.1 Issues in the system

The existing systems present several significant limitations:

1. Fragmented User Experience:

Users must navigate multiple platforms to discover, evaluate, and acquire books, leading to inefficient workflows and poor user satisfaction.

2. Limited Personalization:

Most systems rely on basic search functionality rather than intelligent recommendation algorithms, resulting in poor discovery experiences.

3. No Peer-to-Peer Trading:

Existing platforms lack integrated trading capabilities, forcing users to rely on external services for book exchanges.

4. Academic Community Disconnect:

Traditional systems fail to address the specific needs of university students, including department-based organization and academic calendar considerations.

5. **Mobile Optimization Issues:**

Many existing solutions provide poor mobile experiences, limiting accessibility for users who primarily interact through smartphones.

6. Data Silos:

Information about books, user preferences, and trading opportunities exists in separate systems, preventing comprehensive user profiling and recommendation accuracy.

3.1.2 Solutions

The BookIQ platform addresses these issues through a comprehensive approach:

1. Unified Platform:

Integrates book discovery, recommendation, and trading functionalities into a single, cohesive application that provides a seamless user experience.

2. AI-Powered Recommendations:

Implements sophisticated machine learning algorithms that analyze book content and user preferences to provide personalized suggestions, improving discovery rates and user engagement.

3. Integrated Trading System:

Provides built-in peer-to-peer trading capabilities with secure transaction management, user verification, and status tracking, eliminating the need for external trading services.

4. Academic Community Focus:

Incorporates university-specific features including department categorization, subject-based filtering, and academic calendar integration to serve the unique needs of student users.

5. Mobile-First Design:

Optimizes all functionalities for mobile devices using Flutter's cross-platform capabilities, ensuring consistent performance and usability across Android and iOS platforms.

6.Real-Time Data Synchronization:

Utilizes Supabase's real-time capabilities to maintain data consistency across all platform components, enabling immediate updates and notifications for trading activities.

CHAPTER 4

SYSTEM REQUIREMENTS

4.1 SOFTWARE REQUIREMENTS

Frontend Requirements:

- **Flutter SDK:** Version 3.0.0 or higher
- **Dart SDK**: Compatible with Flutter version
- **Development Environment:** Android Studio, VS Code, or IntelliJ IDEA
- **Platform Support**: Android API level 21+ (Android 5.0+), iOS 11.0+

Backend Requirements:

- **Supabase:** Backend-as-a-Service platform for database, authentication, and real-time **features**
- **PostgreSQL:** Database management system (managed by Supabase)
- **Flask:** Python web framework for machine learning API
- **Python:** Version 3.7 or higher
- **scikit-learn:** Machine learning library for recommendation algorithms

Python Dependencies:

- **flask -** Web framework
- **pandas -** Data manipulation
- **scikit-learn** - Machine learning algorithm**s**
- **numpy -** Numerical computing

Development Tools:

- **Version Contro**l: Git for source code management
- **API Testing:** Postman or similar tools for backend testing
- **Database Management:** Supabase dashboard for database administration
- **Build Tools:** Flutter build tools for application compilation

CHAPTER 5

MODULES

5.1 Authentication Module

- User registration and login functionality
- Email verification system
- Password reset capabilities
- Profile management
- Session management and security
-

5.2 Book Management Module

- Book cataloging and metadata management
- Image upload and storage
- Category and condition tracking
- Availability status management
- Search and filtering capabilities

5.3 Recommendation Engine Module

- TF-IDF vectorization processing
- Cosine similarity calculations
- Content-based filtering algorithms
- Real-time recommendation generation
- Fallback mechanisms for offline mode

5.3 Trading System Module

- Request creation and management
- Status tracking (pending, approved, rejected)
- Communication between users
- Transaction history
- Notification system

5.4 User Interface Module

- Responsive design components
- Dark theme implementation
- Navigation and routing
- State management
- Error handling and user feedback

5.5 Data Synchronization Module

- Real-time data updates
- Offline data caching
- Conflict resolution
- API integration management
- Performance optimization

6. APPENDIX -I

6.1 Code Structure Overview

```
lib/
├── main.dart               # Application entry point
├── models/
│   ├── book.dart           # Book data model
│   └── swap_request.dart   # Trading request model
├── screens/
│   ├── auth/               # Authentication screens
│   ├── home_screen.dart    # Main application screen
│   ├── book_detail_screen.dart
│   └── profile/            # User profile screens
├── services/
│   ├── auth_service.dart   # Authentication logic
│   ├── api_service.dart    # API communication
│   └── email_service.dart  # Email functionality
├── widgets/
│   ├── book_card.dart      # Book display component
│   └── loading_widget.dart # Loading indicators
└── utils/                  # Utility functions
```

6.2 Database Schema

sql

```
-- Users table
CREATE TABLE users (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  email TEXT UNIQUE NOT NULL,
  name TEXT NOT NULL,
  phone_number TEXT,
  university TEXT,
  department TEXT,
  created_at TIMESTAMPTZ DEFAULT NOW(),
  updated_at TIMESTAMPTZ DEFAULT NOW()
);

-- Books table
CREATE TABLE books (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  title TEXT NOT NULL,
  author TEXT NOT NULL,
```

```
  description TEXT,
  owner_id UUID REFERENCES users(id),
  category TEXT CHECK (category IN ('textbook', 'novel', 'reference', 'other')),
  condition TEXT CHECK (condition IN ('excellent', 'good', 'fair', 'poor')),
  price DECIMAL(10,2) NOT NULL,
  is_available BOOLEAN DEFAULT true,
  created_at TIMESTAMPTZ DEFAULT NOW()
);

-- Requests table
CREATE TABLE requests (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  book_id UUID REFERENCES books(id),
  requester_id UUID REFERENCES users(id),
  owner_id UUID REFERENCES users(id),
  type TEXT CHECK (type IN ('swap', 'buy')),
  status TEXT CHECK (status IN ('pending', 'approved', 'rejected', 'cancelled')),
  message TEXT,
  offer_price DECIMAL(10,2),
  created_at TIMESTAMPTZ DEFAULT NOW()
);
```

7. APPENDIX - II

7.1 API Endpoints Documentation

Recommendation API

- **Endpoint:** GET /recommend
- **Parameters:**
- **title:** Book title to find recommendations for
- **top_n:** Number of recommendations (default: 5)
- **Response:** JSON object with input book and recommendation list

Authentication Endpoints

- **Sign Up:** POST /auth/v1/signup
- **Sign In:** POST /auth/v1/token
- **Sign Out:** POST /auth/v1/logout
- **Password Reset:** POST /auth/v1/recover

Book Management Endpoints

- **Get Books:** GET /rest/v1/books
- **Add Book:** POST /rest/v1/books
- **Update Book:** PATCH /rest/v1/books?id=eq.{id}
- **Delete Book:** DELETE /rest/v1/books?id=eq.{id}

7.2 Machine Learning Algorithm Details

The recommendation system uses the following approach:

1. Data Preprocessing:
Combine book title, author, genre, and description into a single text **corpus**

2. TF-IDF Vectorization:
Convert text data into numerical vectors using scikit-learn's TfidfVectorizer

3. **Similarity Calculation:**
Compute cosine similarity between book vectors

4. Recommendation Generation:
Select top N most similar books based on similarity scores

7.3 Security Implementation

- Row Level Security (RLS) policies for data access control
- JWT token-based authentication
- Input validation and sanitization
- Secure API key management
- HTTPS encryption for all communications

8. CONCLUSION REFERENCES

Conclusion

The BookIQ project successfully demonstrates the integration of artificial intelligence-powered recommendation systems with peer-to-peer trading capabilities in a mobile application. The platform addresses key limitations in existing book discovery and trading systems by providing a unified, mobile-optimized solution that serves the specific needs of university students and book enthusiasts.

The implementation of TF-IDF vectorization and cosine similarity algorithms provides effective book recommendations, while the Supabase backend ensures real-time data synchronization and secure transaction management. The Flutter framework enables consistent cross-platform performance and maintains native-like user experiences.

Future enhancements could include advanced machine learning models, social features for book clubs, integration with external book databases, and enhanced analytics for user behavior insights.

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